Computer Science & DA

Warehousing

Lecture - 07

By- Vishvadeep Gothi sir

Recap of Previous Lecture

Topic Data Normalization

Topic Data Aggregation

Topic Data Discretization

Topics to be Covered

- Topic: Data Discretization
- Topic: Clustering
- Topic: Data Generalization
- Topic: Concept Hierarchy

Topic : Data Discretization: Cluster Analysis

Unsupervised

- A clustering algorithm can be applied to discretize a numeric attribute, A, by partitioning the values of A into clusters or groups based on similarity, and store cluster representation (e.g., centroid and diameter) only.
- It partitions the data set into clusters.
- Clustering algorithms
  - K-Means
  - K-Medoid algorithm
- Properties of clusters:

  (i) All the data points in a cluster should be similar to each other.

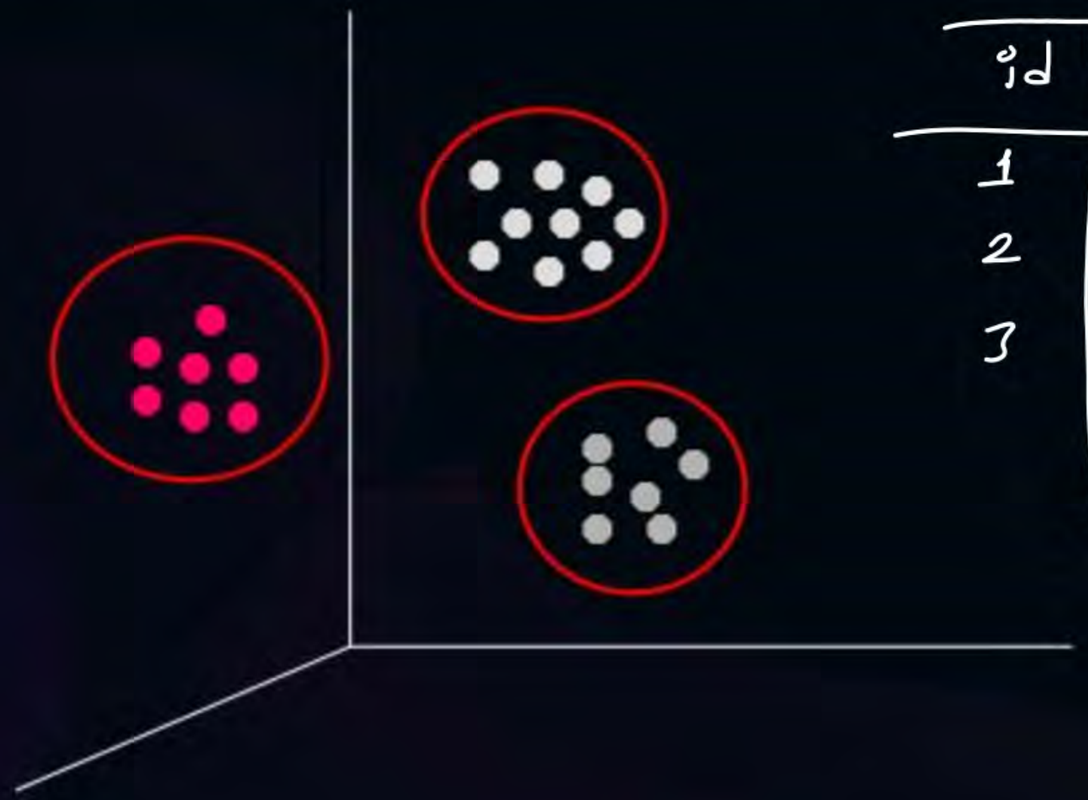
  (ii) The data points from different clusters should be as different as possible.

Topic : Data Discretization: Correlation Analysis

- Supervised discretization method, i.e it uses class information.
- It uses bottom-up merge, i.e. it finds the best neighboring intervals (those having similar distributions of classes) to merge.
- It is performed recursively by finding best neighboring intervals that have a similar distribution of classes and merge them, until a predefined stopping condition

Topic : Clustering

Finding groups of objects such that the objects in a group will be similar (or related) to one another and different from (or unrelated to) the objects in other groups

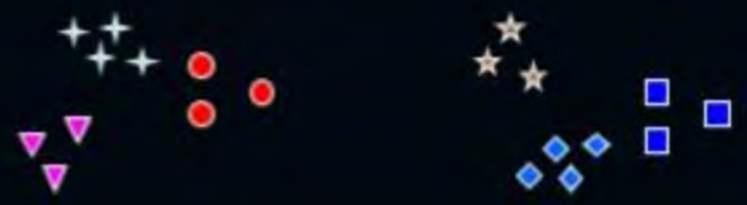


student

id	age	fees
1	20	2000
2	19	2100
3	19	2000

Topic : Clustering Can be Ambiguous

How many clusters?



Six Clusters

Two Clusters

Four Clusters

Topic : Types of Clustering

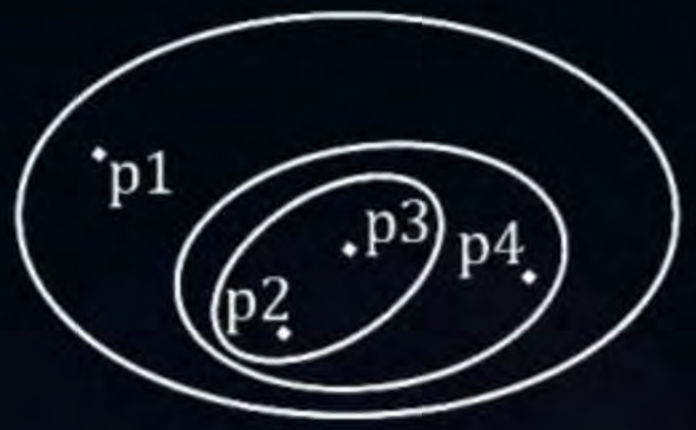
- Partitional Clustering → all data points are clustered in single go.
- Hierarchical clustering → clusters are identified hierarchically

Topic : Partitional Clustering

Original Points

A Partitional Clustering

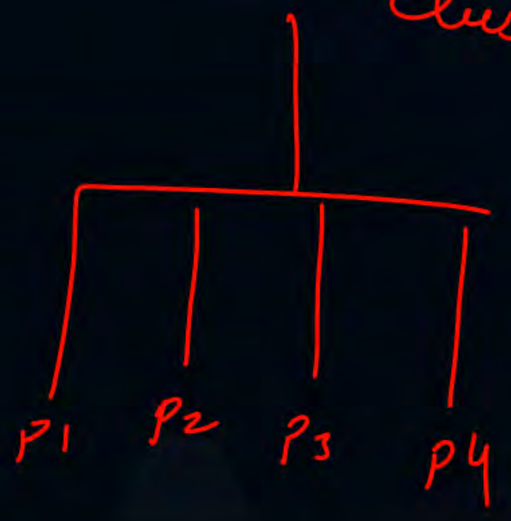
Topic : Hierarchical Clustering



Hierarchical Clustering

Dendrogram

Partitional clustering



Topic : Types of Clusters

- Well-separated clusters
- Center-based clusters
- Contiguous clusters
- Density-based clusters
- Property or Conceptual
- Described by an Objective Function

Topic : Well-separated clusters

3 well-separated clusters

Topic : Center-based clusters

The center of a cluster is often a centroid, the average of all the points in the cluster, or a medoid, the most "representative" point of a cluster

Topic : Continuity-Based clusters

Contiguous Cluster (Nearest neighbor or Transitive)

- Two objects are connected only if they are within a specified distance of each other
- Each point is closer to at least one point in its cluster than to any point in another cluster

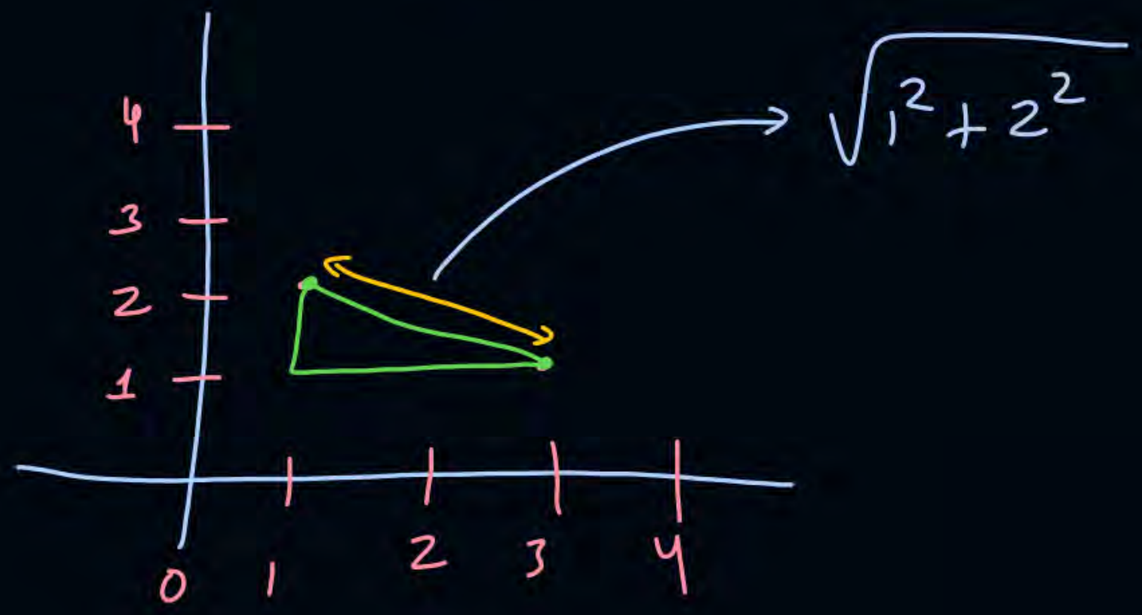
Distance:-

$P_1 \; (x_1, y_1) = (1, 2)$

$P_2 \; (x_2, y_2) = (3, 1)$

distance b/w P_1 and P_2

$$= \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$



Topic : Density-Based clusters

- A cluster is a dense region of points, which is separated by low-density regions, from other regions of high density
- Clusters are regions of high density separated by regions of low density

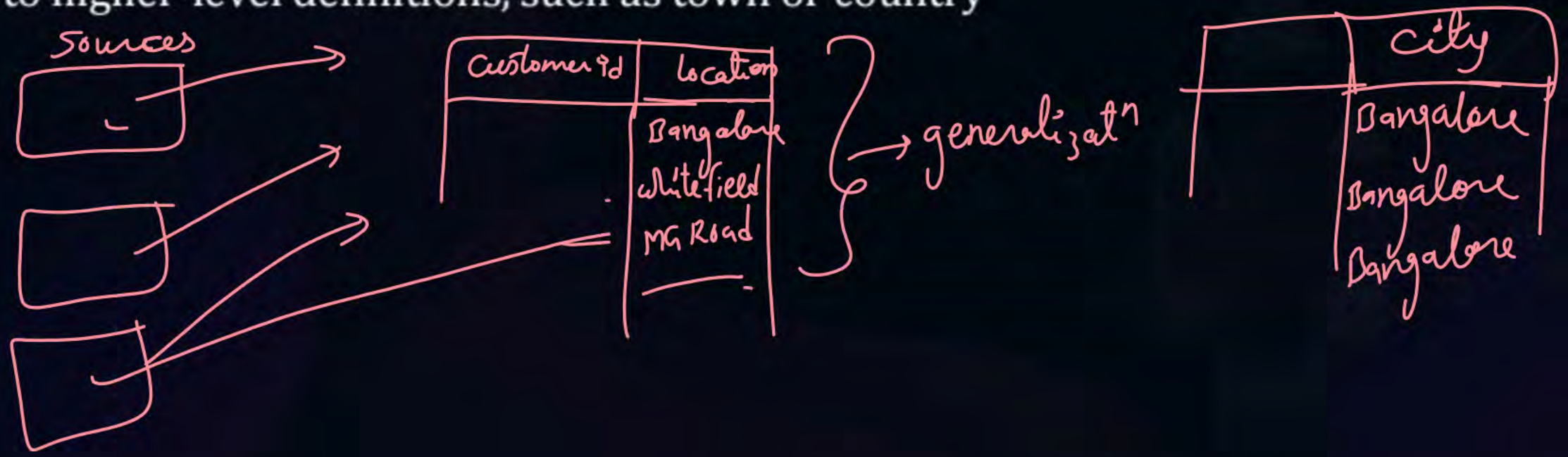
Topic : Property or Conceptual

- Finds clusters that share some common property or represent a particular concept

Topic : Data Generalization

- It converts low-level data attributes to high-level data attributes using concept hierarchy

- Example: Categorical attributes, such as house addresses, may be generalized to higher-level definitions, such as town or country



Topic : Concept Hierarchy

- Concept hierarchy generation is a process of generating hierarchies of concepts based on the data values.

- It refers to the organization of data into a tree-like structure, where each level of the hierarchy represents a concept that is more general than the level below it.

- Many discretization techniques can be applied recursively in order to provide a hierarchical or multiresolution partitioning of the attribute values known as concept hierarchy.

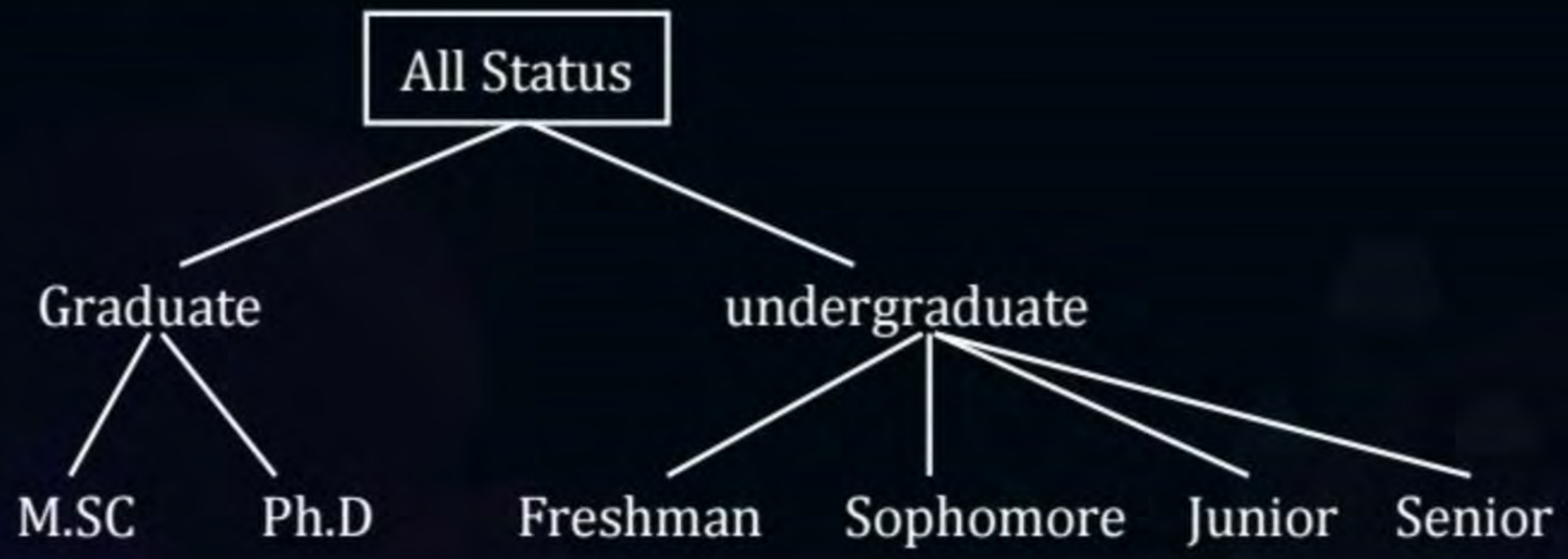
Topic : Concept Hierarchy

- A concept hierarchy for a given numeric attribute attribute defines a discretization of the attribute.

- Concept hierarchies can be used to reduce the data y collecting and replacing low-level concepts (such as numeric value for the attribute age) by higher level concepts (such as young, middle-aged, or senior). Although detail is lost by such generalization, it becomes meaningful and it is easier to interpret.

Topic : Concept Hierarchy

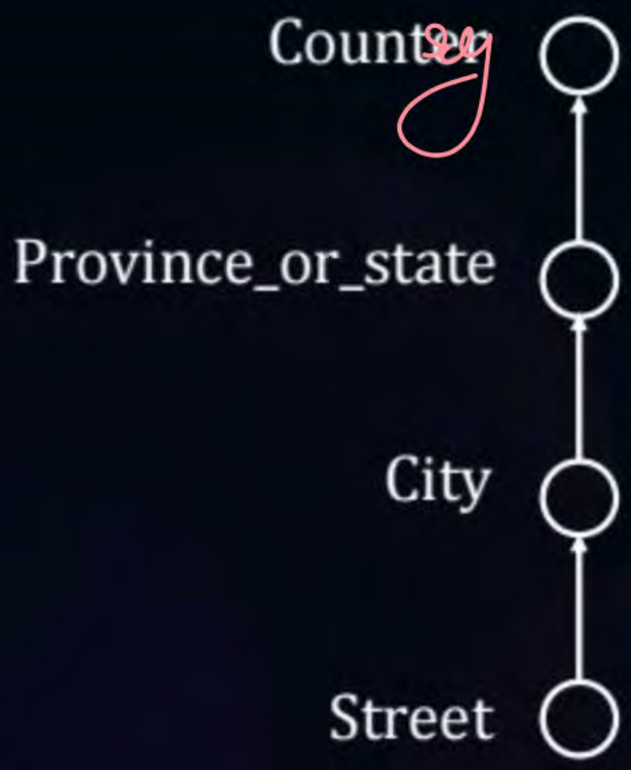
- A set-grouping hierarchy for attribute status



Topic : Concept Hierarchy

- A concept hierarchy for attribute Location

Topic : Concept Hierarchy



Topic : Types of Concept Hierarchy

1. Schema Hierarchy
2. Set-Grouping Hierarchy
3. Operation-Derived Hierarchy
4. Rule-based Hierarchy

Topic : Types of Concept Hierarchy

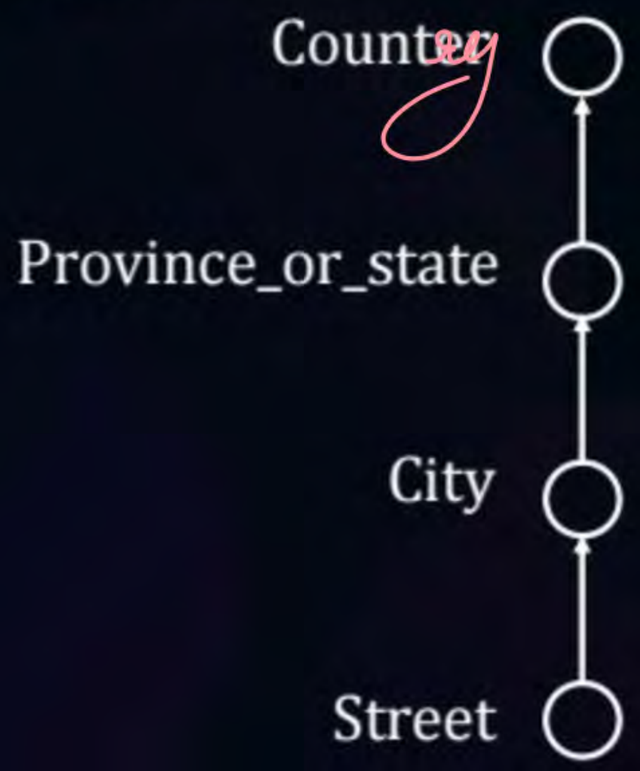
Schema Hierarchy: → for creating schema while creating warehouse

This kind of hierarchy is formed at the schema level by defining the partial order to reflect relationships among the attributes in a database.

Topic : Types of Concept Hierarchy

Schema Hierarchy:

This kind of hierarchy is formed at the schema level by defining the partial order to reflect relationships among the attributes in a database.



Topic : Types of Concept Hierarchy

Set-Grouping Hierarchy:

This kind of hierarchy is formed by defining a set grouping relationship for a set of concepts (or values of attributes) in order to reflect semantic relationships characteristics to the given application domain.

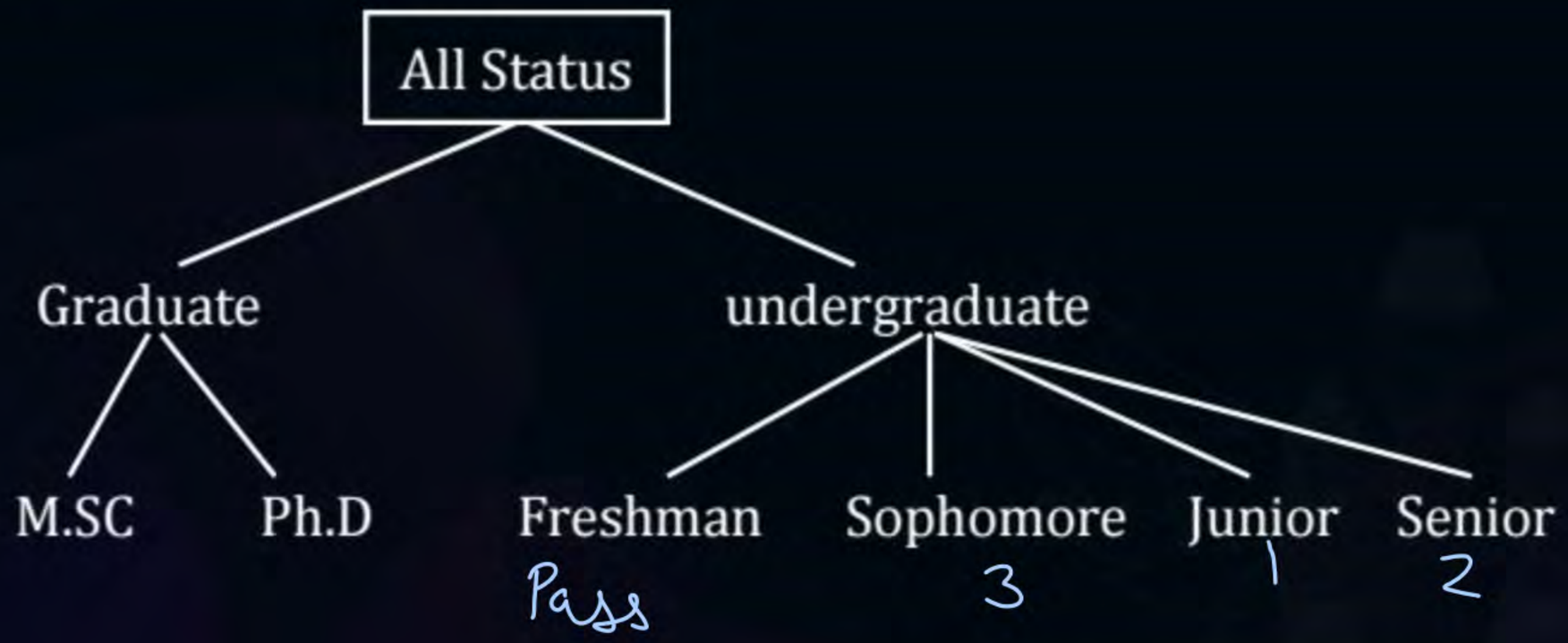
- Also known as instance hierarchy because defined on set of instances or values of an attribute.

ex:-

age	age group	age group
20	20-30	young 20-40
90	90-100	Senior 70-80, 80-90, 90-100
60	60-70	
62	60-70	
53	50-60	
48	40-50	
82	80-90	
26	20-30	
27	20-30	
39		

Topic : Types of Concept Hierarchy

Set-Grouping Hierarchy:



Topic : Types of Concept Hierarchy

Operation-Derived Hierarchy:

Used to organize data by applying a series of operations or transformations to the data. The operations are applied in a top-down fashion, with each level of the hierarchy representing a more general or abstract view of the data than the level below it.

Used in data mining tasks such as clustering and dimensionality reduction

Topic : Types of Concept Hierarchy

Rule-Based Hierarchy:

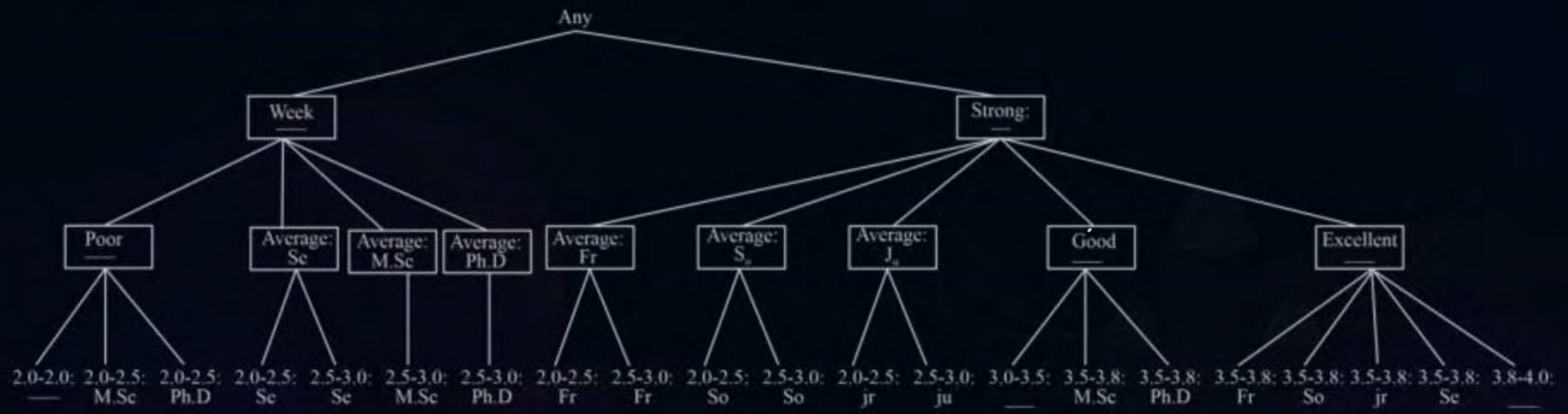
Rule-based Hierarchy is a type of concept hierarchy that is used to organize data by applying a set of rules or conditions to the data.

GPA(0-5) to grade('Excellent', 'Good', etc)

3.6 is good or excellent? → decided based on college, graduation level etc.

Topic : Types of Concept Hierarchy

Rule-Based Hierarchy:



2 mins Summary

Topic Data Discretization

Topic Clustering

Topic Data Generalization

Topic Concept Hierarchy

Happy Learning

THANK - YOU